

Lesson Exemplar in General Science

Quarter 2

LE

2

Lesson Exemplar for General Science
Quarter 2: Unit 2

This material is intended exclusively for the use of senior high school teachers participating in the implementation of the Strengthened Senior High School Curriculum. It aims to assist in delivering the curriculum content, standards, and lesson competencies. Any unauthorized reproduction, distribution, modification, or utilization of this material beyond the designated scope is strictly prohibited and may result in appropriate legal actions and disciplinary measures.

Borrowed content included in this material are owned by their respective copyright holders. Every effort has been made to locate and obtain permission to use these materials from their respective copyright owners. The publisher and development team do not represent nor claim ownership over them.

Development Team	
Writer:	Cherry V. Abrazado PhD Region I Division of Alaminos City
Validator:	Catherine O. Sison Region XII
Language Editor:	
Consultant:	Alfons Jayson O. Pelgone PhD Philippine Normal University, Manila
Learning Area Specialist:	John Cyrus L. Doblada
Bureau of Learning Delivery Bureau of Curriculum Development Bureau of Learning Resources	

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph.

LESSON EXEMPLAR

Learning Area	GENERAL SCIENCE	Grade Level	11
Semester	FIRST	Quarter	2

I. OBJECTIVES *(Identifying the Goals)*

Content Standard	<i>The learners demonstrate understanding that household and personal care products contain chemical substances which determine their properties and guide their proper use and disposal.</i>
Performance Standard	<i>The learners apply research skills to explore how scientists have made an impact on our lives. They recognize the significant role of different agencies in the Philippines, such as DOST and DTI, in driving scientific innovation and development. They demonstrate critical thinking by explaining the various substances present in household and personal care products, including their benefits and the potential risks associated with their use and disposal. They practice appropriate and safe handling of chemicals. They also explain that chemical reactions occur in human activities and in the environment, and that several chemical solutions may pose threats to the community.</i>
Learning Competencies	Key Topics Learned in Previous Grade Level 1. Describe the chemical properties of materials, such as they can be burnt, react with other materials, or are degradable or biodegradable 2. Identify that matter has (exists in) three states called solids, liquids, and gases 3. Describe useful everyday examples of uniform and non-uniform mixtures, such as solutions and suspensions 4. Recognize the symbols of common elements and the formula for common compounds. 5. Describe the properties of ionic, covalent, and metallic substances. 6. Explain how some important chemical reactions impact the natural and built environments.

	<i>In this lesson, Grade 11 learners will be able to:</i> <i>Construct a table using product labels that describes the function of the different substances (elements or compounds) present in household products such as those in bleach, detergents, baking powder and personal care products such as shampoo, soaps, toothpaste.</i>
II. REFERENCES and MATERIALS <i>(Selecting Resources and Material)</i>	• Fashionlady. "Understanding What Is Cosmetic Chemistry: Why Is It So Important to Know Before Applying Makeup?" Fashionlady, March 13, 2016. • https://www.fashionlady.in/understanding-what-is-cosmetic-chemistry-why-is-it-so-important-to-know-before-applying-makeup/65382 • IvyRose. 2019. Common Chemicals Found in the Home: Basic Chemistry. Accessed November 7, 2019. http://www.ivyroses.com/Chemistry/GCSE/Common-Chemicals-in-the-Home_Molecular-Formulae.php • Writers, Starts at 60. 2019. Six Things You Can Use Instead of Bleach When Cleaning. Starts at 60, October 18. https://starts60.com/discover/lifestyle/6-things-you-can-use-instead-of-bleach-when-cleaning • Prezi. 2025. 10 Household Products & Their Active Ingredients. Accessed May 7. https://prezi.com/oxydtyhideiq/10-household-products-their-active-ingredients/. • Solubility of Things. Chemistry in Household Products. Accessed May 7, 2025. https://www.solubilityofthings.com/chemistry-household-products
III. CONTENT <i>(Sequencing Content)</i>	<i>Application of Chemical Substances in Household and Personal Care Products</i> • Composition of household and personal care products. • Physical and Chemical Properties of household and personal care products • Product labels of household and personal care products • Function of substances found in household and personal care products using product label
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	<i>Unpacked Learning Competencies</i> 1. Identify the substances in common household products such as those in bleach, detergents, baking powder and personal care products such as shampoo, soap, and toothpaste. 2. Describe the function of each substance present in household products such as those in bleach, detergents, baking powder and personal care products such as shampoo, soap, and oothpaste.

	3. <i>Justify a decision on consumer choices based on composition of the household product and personal care products listed in product label.</i>													
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i>		ANNOTATION <i>*Instruction to teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>												
A. Activating Prior Knowledge	DAY 1 1. Activating Prior Knowledge <i>Household and personal care products play a vital role in maintaining cleanliness, hygiene, and overall well-being. From detergents and disinfectants that keep our spaces germ-free to shampoos and skincare essentials that support self-care, these products are designed to improve daily living while ensuring safety and comfort. Today, we will dive into this topic together. Let's start our activity!</i> Option 1: Knowledge Quest <i>The teacher will ask the students to create three columns labeled: K – What do I KNOW? W – What do I WANT to know? H – How do I FIND OUT? The learners are encouraged to fill out all the three columns about household and personal care products</i> <table border="1"> <thead> <tr> <th>K</th><th>W</th><th>H</th></tr> </thead> <tbody> <tr> <td>What I Know?</td><td>What I Want to know?</td><td>How do I find out?</td></tr> <tr> <td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td></tr> </tbody> </table>	K	W	H	What I Know?	What I Want to know?	How do I find out?							<i>* The teacher will choose one activity from three options. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback</i> <i>* The IDF elements present in this section are reflective, in nature. It is reflective because the activity focuses on critical thinking, drawing from related experiences.</i> <i>It is in the Analysis phase because it involves identifying learning goals, understanding learners' needs, analyzing tasks, and setting clear instructional objectives</i> <i>This is ideational because it emphasizes the things or characteristics that students see and</i>
K	W	H												
What I Know?	What I Want to know?	How do I find out?												

	Option 2: Picture Classification of Common Consumer Product in household <i>The teacher will present random images of common consumer products. Learners will carefully observe and analyze each image, then classify the item into one of the following categories: household product or a personal care product. The learners must justify their choice by explaining what the product is used for and how it fits into the selected category. They may refer to its function, ingredients, or common usage in households. To encourage active participation, learners can work individually, in pairs, or in groups, discussing their reasoning before sharing with the class. Learners answer the guide questions.</i> Option 3: ABC of Home Essentials <i>The teacher prepares cut-out letters representing all 26 letters of the alphabet and places them in a container. The teacher randomly draws a letter from the container. The chosen student or group must quickly think of a household product that starts with the selected letter. They must then explain its purpose and function, highlighting how it is used in daily life. Learners can work individually, in pairs, or in teams, ensuring collaboration and engagement.</i> Guide Questions: <i>1. What is your basis for identifying a household product from personal care products?</i> <i>2. What do you think is the importance of household and personal care products in our daily lives? Why?</i>	<i>experience (context) in order perform the activity.</i> <i>The guide questions also serve as a formative assessment, allowing the teacher to evaluate the breadth and depth of students' understanding of the concept</i> <i>This belongs to the engage phase, where learners actively connect with the topic, explore ideas, and build curiosity before diving deeper into the lesson.</i>
--	--	---

	2. Establishing the Purpose of the Lesson <i>Through our previous activity, we have seen how consumer products play a vital role in our daily lives. Now, we will explore that understanding household and personal care products composition and functions help us make informed choices that protect our health, improve hygiene, and contribute to a safer environment.</i> Option 1. STEAM Vocabulary: Understanding Key Terms <i>The teacher will instruct the students to form several groups. Each group will be provided a list of words related to household and personal care products. Each group will do the following:</i> <i>1. Sort the given words into STEAM categories (Science, Technology, Engineering, Arts and Mathematics)</i> <i>2. Define each word and explain why it's fits in a specific STEAM category</i> <i>3. Find examples of how the term is used in real-world contexts, especially in household and personal care products.</i> <i>4. Share the output in the class.</i> <i>*Sample word: surfactants</i> <i>1. STEAM Categories</i> <i>Surfactants-Science</i> <i>2. Definition</i> <i>Surfactants -Chemicals that lower surface tension, making cleaning products effective for removing dirt and grease.</i> <i>3. Real-world Example</i> <i>Surfactants help dishwashing liquids cut through grease</i>	<i>* The teacher will choose one activity from two options. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>* This section is both responsive and reflective as it recognizes the diverse backgrounds of students while emphasizing critical thinking and metacognition. It fosters an inclusive learning environment, encouraging students to analyze concepts deeply and connect their experiences to broader ideas.</i> <i>The design phase focuses on sequencing content, selecting appropriate strategies, and making learning meaningful which ensures coherence and engagement</i>
--	---	---

	Option 2: Spot, Observe, and Learn <i>The teacher encourages the learners to present the product labels. Learners take turns to examine the ingredients listed on the labels, identifying the active substances. They compare labels, noting common ingredients across different products. Learners will answer the guide questions.</i> Guide Questions: <i>1. Are there similarities based on the ingredients among the household products?</i> <i>2. What are the differences among these products based on the ingredients listed on the labels?</i>	<i>*The teacher asks the learners in advance to bring collected product labels from their homes</i>
B. Instituting New Knowledge	DAY 2 1. Presenting Examples <i>As we move forward, let's connect what we see and use daily to the science behind their ingredients and their role in our lives.</i> Option 1. Mind mapping: Product Connections <i>The teacher will ask the students to create a mind map to visually organize and explore the functions of various household and personal care products. Learners can choose a central topic either household or personal care products. Then expand with subcategories (Shampoo" under Hygiene, "Detergent" under Cleaning). They can be also guided to identify Key connections by drawing a link between products that have similar (e.g., "Moisturizing" connects lotion and conditioner). Learners will encourage to share their mind maps with the class.</i> <i>*Note: Learners can also do color-coded Mind Map- Assign colors based on product type, use, or ingredients, and effects.</i>	<i>* The teacher will choose one activity from two options given. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>* This remains a formative assessment, designed to provide insight into students' understanding. It is integrative, as it takes a broad perspective on concepts, considering various aspects. It is also linked to context and connection, focusing on students' backgrounds and related experiences. Furthermore, it falls under the explore phase, as the activity allows students to navigate and deepen their knowledge.</i>

	Option 2: Discuss-Reflect-Connect-Understanding Household and Personal Care Products <i>The teacher will group the class into five (5). The members of the group will collaborate and brainstorm on common household and personal care products found in their home. They will talk about function and the properties of the ingredients of the products. They will share their findings with the class.</i> Guide Questions: <i>1. Are certain ingredients found in multiple products? What does this suggest about their use?</i> <i>2. What surprised you the most after doing the activity on understanding household and personal care products?</i> 2. Discussing New Concept <i>Through our previous activity, we have seen how consumer products have active ingredients that distinguish them from other household and personal care products. Now, we will explore how understanding the properties and function of household and personal care products and using them allows us to make informed choices that promote health, hygiene, and environmental responsibility.</i> Option 1: Research and Present! <i>The teacher organizes the learners into groups beforehand and assigns them to research the classification of household products (e.g., cleaning agents, personal care products, air and fabric fresheners, fabric softeners, and pest control products), including their ingredients and functions. The learners then share their findings, which may be presented through a PowerPoint presentation. At the end of the presentation, the teacher will provide additional details and supporting</i>	<i>* The teacher will choose one activity from two activities. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>This approach is inclusive, ideational, and integrative. The processing questions acknowledge various contexts, facilitate conceptual understanding, and provide a holistic perspective. It also emphasizes context and connection, ensuring that learners relate their experiences to the discussion.</i>
--	--	--

data to clarify misconceptions and further explanations. Learners will take note the important concepts then will answer the guide questions

Option 2: Visual Learning: Exploring Household & Personal Care Products

The teacher will play a video about the function and the substances in household and personal care products followed by a short discussion facilitated by the teacher. Learners will take note of the important concepts and answer the guide questions. (Source: [The Chemistry of Everyday Life: Household Chemicals Explained](#))

Guide Questions:

1. What role do different substances play in making household and personal care products effective?
2. What similarities and differences exist in the substances found in various personal care and household products?

Chemistry Behind common Household and Personal Care Products

Cleaning Products and their Key Components

Category	Function/ Purpose	Active Ingredient(s)	Key Components and Roles
Bleach	Breaks down stains and kills bacteria through oxidation.	Sodium Hypochlorite (NaClO)	Calcium Hypochlorite (Ca(ClO) ₂) -water purification. Chlorine Gas (Cl ₂) industrial bleaching. Chlorine Dioxide (ClO ₂) – Strong oxidizing agent.
Detergent	Removes dirt, grease, and stains from	Surfactants and Enzymes	Surfactants – lifts dirt and grease. Enzymes – break down protein-based stains.

*. The teacher can add more additional substances found in common household and personal care products.

Note: Household products and personal care products will be discussed separately to help learners distinguish their functions and ingredients, making the lesson easier to understand.

		<i>fabrics, dishes, and surfaces.</i>		<i>Builders – softens water and enhances cleaning power. Bleaching Agents – stain removal and disinfection. - Fragrances and Dyes – adds scent and color.</i>
	<i>Baking Soda</i>	<i>Acts as a leavening agent, odor neutralizer, and mild abrasive for cleaning.</i>	<i>Sodium Bicarbonate (NaHCO₃)</i>	<i>Carbon Dioxide (CO₂) – released in reactions, creating bubbles in baked goods.</i>
	<i>Degreaser</i>	<i>Removes grease, oil, and stubborn residues from surfaces.</i>	<i>Solvents and Alkaline Builders</i>	<i>Surfactants – emulsify and disperse grease. Solvents – dissolve grease deposits. Alkaline Builders – adjust pH for better cleaning. Chelating Agents –prevent residue buildup. Corrosion Inhibitors -protect metal surfaces.</i>
	<i>Air Freshener</i>	<i>Neutralizes odors and releases pleasant scents indoors.</i>	<i>Fragrances and Odor Neutralizers</i>	<i>Essential Oils provide scent. Propellants -helps disperse fragrance. Odor Neutralizers –absorb unwanted smells. Volatile Organic-Compounds (VOCs) –found in some sprays, may affect air quality.</i>
	<i>Fabric Softeners</i>	<i>Reduces static cling, softens fabrics, and adds fragrance to laundry.</i>	<i>Cationic Surfactants</i>	<i>Fragrances – provides a fresh scent. Silicones and Conditioning Agents – smooths fabric texture.</i>

			Preservatives and Stabilizers – maintains product effectiveness over time.
--	--	--	---

Day 3

Personal Care Products and their Key Components

Category	Function/ Purpose	Active Ingredient(s)	Key Components and Roles
Shampoo	Cleanses, nourishes, and protects hair and scalp.	Surfactants (Sodium Lauryl Sulfate, Ammonium Laurate Sulfate)	Water – main base ingredient. Conditioning Agents – softens and detangles (Silicones, Fatty Alcohols). Preservatives – prevent bacterial growth (Parabens, Benzyl Alcohol). Fragrances and Additives – adds scent and UV protection.
Soap	Removes dirt, oil, and bacteria; hydrates and protects skin.	Sodium Hydroxide (NaOH) / Potassium Hydroxide (KOH)	Fatty Acids and Oils – base ingredients (Coconut Oil, Olive Oil). Glycerin – retains moisture. Fragrances and Essential Oils – add scent and skin benefits. Antibacterial Agents – helps fight germs (Triclosan).
Toothpaste	Cleans, protects, and maintains oral health.	Fluoride (Sodium Fluoride, Stannous Fluoride)	Abrasives – removes plaque (Calcium Carbonate, Hydrated Silica). Humectants – prevents drying out (Glycerin, Sorbitol). Detergents – create foam (Sodium Lauryl Sulfate). Antibacterial Agents – prevents gum disease and bad breath (Triclosan, Stannous Fluoride). Flavoring Agents – enhance taste (Mint, Sweeteners).

	3. Developing Mastery <i>Take a moment to reflect on what you know about household and personal care products—how they work, what substances they contain, and why they are essential in daily life. With this knowledge, let's explore their functions through experimentation!</i> Option1: Investigating the Cleaning Power of Household and Personal Care Products <i>The teacher will instruct the students to perform an experiment to compare the effectiveness of different cleaning agents found in household and personal care products and how their substances work. Learners will be instructed to apply an equal amount of oil or food stain to different surfaces, then add a small amount of each cleaning product to separate stained areas. They will scrub each section for the same duration, then rinse with water. Learners compare the stain removal effectiveness. The learners will test the pH level of each product to understand its chemical properties.</i> Materials: <i>Three types of cleaning agents (e.g., soap, shampoo, detergent, toothpaste), water, cotton fabric or white tiles (as test surfaces), cooking oil or a food-based stain (e.g., ketchup or coffee), small brushes or sponges, measuring cups, pH strips</i> Guide Questions <i>1. Which product removes the stain most effectively?</i> <i>2. Which cleaning product was the most effective?</i> <i>3. What conclusions can you draw about the role of chemical properties in cleaning products?</i>	<i>* The teacher will choose one activity from two options. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides a constructive feedback.</i> <i>*This approach is relevant and responsive, incorporating transdisciplinary learning and problem-solving. It transcends language and the formation of ideas, connecting with various fields of knowledge. Additionally, it is solution-oriented, focusing on addressing challenges.</i> <i>Furthermore, because it nurtures multiple skills and areas of expertise, it is integrative—encouraging learners to apply diverse knowledge. The process also fosters collaboration and creativity.</i>
--	--	--

	Option 2: Decoding Product Labels: Understanding What We Use Daily <i>The teacher instructs the learners to work individually. Learners will examine the product labels, focusing on ingredients, warnings, and marketing claims. They will identify active substances and their possible effects. Their output will be shared with the class.</i> Guide Questions: <i>1. Can you identify the active substances and their function using the product label?</i> <i>2. Does the product label include any warnings or precautions?</i> <i>3. How can reading product labels help us make better choices?</i>	
C. Demonstrating Knowledge and Skills	Day 4 1. Finding Practical Application <i>Now that we've explored the functions and key substances in household and personal care products, let's take our learning beyond the classroom. How can we apply this knowledge to real-life situations?</i> Option 1: Analyzing Product label <i>The learners will be instructed to bring various household product labels. The teacher will provide a table and the learners will fill in each column on the product name, key ingredients and the function of each substance using the product labels they bring. They will compare multiple product labels, noting similarities and differences in ingredients. The learners present their completed tables and justify their chosen product.</i> Option 2: Understanding Product Use <i>Students will take on the role of a kitchen staff member in a busy restaurant, responsible for washing dishes and ensuring utensils are properly cleaned. However, the employee has skin sensitivity to strong</i>	<i>*The teacher will choose one activity from two given options. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>This section is both relevant and reflective, as it hones creativity, critical thinking, and analytical skills. It encourages learners to engage deeply with concepts, fostering a thoughtful and insightful approach to understanding.</i>

	detergents, making certain dishwashing products difficult to use. The restaurant is located near their home, making it a practical job choice. Finding another job would require long travel hours, which is not feasible. Given this situation, students must choose and justify a dishwashing product that is both effective in cleaning and safe for sensitive skin, explaining the reasoning behind their selection 2. Making Generalization Now that we have explored the functions and substances in household and personal care products, let us take a moment to reflect on the key takeaways from our lesson Option 1: Connecting the Dots The teacher facilitates the activity by asking the students to write three key takeaways from the lesson on the substance, function of the common household and personal care products on sticky notes and post them on the board, and group similar insights together to create a summary statement about the lesson. Option 2: Report Outline Learners research and analyze product ingredients, summarizing key findings, and write an analytical report justifying the selection of a product based on scientific reasoning 3. Evaluating Learning Now that we have explored, analyzed, and applied our learning, it is time to assess how well we've understood and integrated these concepts.	<i>*The teacher will choose one activity from two options. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides a constructive feedback</i> <i>*This is reflective, as it highlights critical thinking and metacognition in the development of the activity and its explanation. Additionally, the empathize stage is present, where students can express their ideas and connect personally with the subject matter.</i>
--	--	---

	Option 1: Modified TRUE or FALSE <i>The teacher will instruct the students to write the word 'TRUE' if the statement is correct. If the statement is incorrect, they must write 'FALSE' and then provide the correct answer by modifying the statement. Answers should be written on another sheet of paper.</i> <i>Dishwashing liquid is primarily used to clean dishes and contains surfactants that help remove grease and food residues.</i> <i>Toothpaste contains fluoride, which helps strengthen teeth and prevent cavities.</i> <i>Bleach is a safe cleaning product for all types of surfaces, including wood and fabric.</i> <i>Shampoo is designed to clean hair and often contains sulfates to create lather and remove oil and dirt.</i> <i>Air fresheners only contain fragrances and essential oils to neutralize odors.</i> <i>Most hand soaps contain glycerin, which helps keep the skin moisturized after washing.</i> <i>Toothpaste contains citric acid to kill bacteria and prevent tooth decay.</i> <i>Fabric softeners contain cationic surfactants, which coat fabric fibers to reduce static cling and make them softer.</i> <i>Preservatives in shampoo, such as parabens and benzyl alcohol, are added to enhance its fragrance</i> <i>0. Volatile Organic Compounds (VOCs) found in some air fresheners improve indoor air quality and are completely safe.</i> 4. Additional Activities Option 1: Household & Personal Care Guide <i>Learners will make a tri-fold brochure about common household and personal care products. Include the different classification, ingredients,</i>	<i>*The teacher will choose one activity from two activities. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides a constructive feedback.</i> <i>*This section is both relevant and reflective, as it develops comprehensive knowledge and skills.</i>
--	--	--

	and their function. They may use English, Filipino and other local languages as the medium. Option 2. Knowledge & Next Steps The teacher asks the learners organize their responses into a structured table with three columns <table border="1"> <thead> <tr> <th>L</th><th>A</th><th>Q</th></tr> </thead> <tbody> <tr> <td>What Have I learned?</td><td>What action will I take?</td><td>What new questions do I have?</td></tr> <tr> <td></td><td></td><td></td></tr> </tbody> </table>	L	A	Q	What Have I learned?	What action will I take?	What new questions do I have?				<i>*Use the KWH activity in the beginning of the lesson if you will utilize the LAQ activity.</i>
L	A	Q									
What Have I learned?	What action will I take?	What new questions do I have?									
V. ASSESSMENT (Assessing Learnings)	A. Read each item carefully and write the letter of the correct answer in a separate sheet of paper. - Which ingredient in bleach is primarily responsible for its disinfecting and whitening effects? A. Calcium hypochlorite ($\text{Ca}(\text{ClO})_2$) B. Sodium hypochlorite ($\text{NaClO}(\text{ClO}_2)$) C. Chlorine dioxide (ClO_2) D. Chlorine gas (Cl_2) - If you need to remove protein-based stains like food or blood, which component in detergents would be most effective? A. Builders B. Enzymes C. Fragrances D. Surfactants - Why are chelating agents important in degreasers? A. They dissolve grease deposits. B. They enhance cleaning power by adjusting pH. C. They protect metal surfaces during cleaning D. They remove metal ions to prevent residue buildup. - What is a potential environmental concern regarding air fresheners that contain Volatile Organic Compounds (VOCs)? A. They neutralize odors effectively. B. They contribute to indoor air pollution. C. They contain antibacterial properties D. They release pleasant scents into the air. - Which household product would be most suitable for neutralizing odors and cleaning surfaces with mild abrasiveness? A. Baking Soda B. Bleach C. Detergent D. Fabric Softener - Which ingredient in shampoo acts as a cleansing agent to remove dirt and oil? A. Conditioning Agents B. Fragrances C. Preservatives D. Surfactants - If a toothpaste is labeled as anti-cavity, which ingredient is most likely responsible for this property? A. Abrasives B. Flavoring Agents C. Fluoride D. Humectants										

8. Why are preservatives included in shampoo?
- A. To remove dirt and oil
B. To soften and detangle hair
C. To provide UV protection
D. To prevent bacterial growth and extend shelf life
9. What is the role of glycerin in soap?
- A. To create foam
B. To provide scent
C. To retain moisture in the skin
D. To act as a disinfecting compound
10. What is a potential concern regarding the use of antibacterial agents in soap?
- A. They improve the scent of the soap.
B. They are ineffective against bacteria.
C. They may cause skin dryness with frequent use
D. They reduce the soap's ability to hydrate the skin.

B. Exploring Product Ingredients & Functions

Learners will construct a table using product labels that describes the function of the different substances (elements or compounds) present in household products. The teacher and learners will agree on assessment criteria, ensuring fairness in evaluation. A rubric-based scoring system will be used to provide transparent and objective feedback on engagement, accuracy, and skill application.

Sample Table

Product name	Ingredients	Function/Purpose	Possible Benefits or Risk
toothpaste	*Fluoride *Calcium Carbonate	Strengthens enamel and prevent cavities. Remove stains and plaque	*Benefits- Prevent cavities, Fight harmful bacteria, and promotes long-term oral health *Risk- Dental Fluorosis, gastrointestinal issues

Sample Rubric

Criteria	Excellent	Proficient	Developing	Needs Improvement
Completeness	Includes all required product names, ingredients, functions, and benefits/risks.	Most entries are complete but may lack details in one category.	Some tasks are missing, affecting overall analysis.	Many entries are incomplete, making the table ineffective
Accuracy of Information	All details are correct, with well-researched function and effects of ingredients	Mostly accurate, with minor errors in descriptions or classifications	Some inaccuracies or unclear definitions present	Several errors, misleading informal or missing explanation
Organization and Presentation	Table is well-structured, clearly labeled, and easy to understand	Mostly organized, but some sections lack clarity	Some disorganization, making it hard to follow	Poorly structured and difficult to interpret

VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	<i>This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.</i>
---	---

Prepared by:

CHERRY V. ABRAZADO PhD

Master Teacher II

Alaminos City National High School

Alaminos City Division/Region I

Approved by:

JOHN CYRUS L. DOBLADA

Learning Area Specialist

BLD-TLD

Reviewed by:

ALFONS JAYSON O. PELGONE PhD

Philippine Normal University

APPENDIX

Activity 1: Investigating the Cleaning Power of Household and Personal Care Products Cleaning Product Effectiveness Analysis

Product Name	Type of Stain Tested (Coffee/ Ink/ Grease)	Active Ingredients	Effectiveness (Strong/ Moderate/ Weak)	Observed Differences in Cleaning Power	pH Level
Baking Soda	To be observed	Sodium Bicarbonate (NaHCO ₃)	To be rated	To be described	To be measured
Bleach	To be observed	Sodium Hypochlorite (NaClO)	To be rated	To be described	To be measured
Detergent	To be observed	Surfactants, Enzymes	To be rated	To be described	To be measured
Shampoo	To be observed	Surfactants, Conditioning Agents	To be rated	To be described	To be measured
Toothpaste	To be observed	Fluoride, Abrasives	To be rated	To be described	To be measured

Modified TRUE or FALSE Activity: Understanding Household and Personal Care Products

1. TRUE – Dishwashing liquid is primarily used to clean dishes and contains surfactants that help remove grease and food residues.
2. TRUE – Toothpaste contains fluoride, which helps strengthen teeth and prevent cavities.
3. FALSE – Bleach is not safe for all surfaces; it can damage wood and fabric.

Corrected Statement: Bleach is a strong cleaning product but can damage wood and fabric surfaces.

4. TRUE – Shampoo is designed to clean hair and often contains sulfates to create lather and remove oil and dirt.
5. FALSE – Air fresheners may also contain other chemicals, not just fragrances and essential oils.

Corrected Statement: Air fresheners contain fragrances, essential oils, and other chemicals that help neutralize odors.

6. TRUE – Most hand soaps contain glycerin, which helps keep the skin moisturized after washing.
7. FALSE – Toothpaste does not typically contain citric acid for killing bacteria and preventing tooth decay.

Corrected Statement: Toothpaste contains fluoride to strengthen enamel and antibacterial agents to help fight bacteria.

8. TRUE – Fabric softeners contain cationic surfactants, which coat fabric fibers to reduce static cling and make them softer.
9. FALSE – Preservatives in shampoo are added to prevent bacterial growth and extend shelf life, not to enhance fragrance.

Corrected Statement: Preservatives in shampoo, such as parabens and benzyl alcohol, are used to prevent bacterial growth and extend shelf life.

10.FALSE – Some Volatile Organic Compounds (VOCs) in air fresheners can negatively impact indoor air quality and may not be completely safe.

Corrected Statement: Some Volatile Organic Compounds (VOCs) in air fresheners can contribute to indoor air pollution and may have health risks with prolonged exposure.

ASSESSMENT

1. B. Sodium hypochlorite (NaClO)

Rationale: Sodium hypochlorite is the primary active ingredient in liquid bleach, responsible for its disinfecting and whitening effects through oxidation.

2.B. Enzymes

Rationale: Enzymes are specifically designed to break down protein-based stains like food and blood, making them effective in stain removal.

3. D. They remove metal ions to prevent residue buildup

Rationale: Chelating agents bind to metal ions, preventing them from interfering with the cleaning process and reducing residue buildup.

4. B. They contribute to indoor air pollution

Rationale: Some air fresheners contain VOCs, which can contribute to indoor air pollution and may pose health risks over time.

5. A. Baking Soda

Rationale: Baking soda is a mild abrasive that neutralizes odors while also helping to clean surfaces effectively.

6. D. Surfactants

Rationale: Surfactants help shampoos remove dirt and oil by reducing surface tension and allowing water to spread and lift residues.

7. C. Fluoride

Rationale: Fluoride helps strengthen enamel and prevent cavities, which is why it is included in anti-cavity toothpaste.

8. D. To prevent bacterial growth and extend shelf life

Rationale: Preservatives are added to shampoo to prevent microbial contamination and ensure the product remains safe for use.

9. C. To retain moisture in the skin

Rationale: Glycerin is a humectant that helps draw moisture to the skin, preventing dryness after washing.

10.C. They may cause skin dryness with frequent use

Rationale: Some antibacterial agents, such as triclosan, can strip natural oils from the skin, leading to dryness with excessive use